Supplementary B. CRAT ablation analysis

This supplementary analysis details the ablation of CRAT's components.

B.1 Comparison basis and scope

The displayed ablation results favor the complete CRAT configuration on OPUS100 across EN-FR, EN-FA, and EN-ZH, but the strength of the observed differences depends on the language pair and evaluation metric. The largest BLEU decreases occur on EN-ZH, whereas the largest chrF and COMET decreases occur on EN-FA. Disabling KL regularization or using static roles produces the largest reductions in training memory and time. Removing the causal-prior loss produces the clearest change in the reported routing patterns.

These findings require a qualification about the reference scores. All 54 supplied quality differences disagree with differences calculated from the displayed Full-CRAT means. The displayed full-model means and standard deviations exactly match the earlier Mistral main-comparison results, while the ablation tables report 2.5 times as many training updates. We therefore report descriptive comparisons against the displayed Full-CRAT rows and retain the reference discrepancy in Appendix A. Recomputing the differences resolves their arithmetic; confirming the underlying checkpoints is still necessary before interpreting them as controlled component effects. The ablation exports do not independently identify a backbone, so these tables cannot establish corresponding effects for both Mistral and Llama 2.

We retain the export labels Full-CRAT, no-kl, no-causal, no-morph, no-sparse, no-integration, and static-role. Their intended contrasts are, respectively, the complete configuration, removal of KL regularization, removal of causal-prior alignment, removal of the auxiliary morphology objective, removal of the soft-gate sparsity penalty, removal of integration adapters, and replacement of dynamic routing with static roles. The earlier implementation supports these loss and architecture concepts, but the main-comparison scripts do not define every exported ablation label. The exact ablation configurations must therefore accompany any definitive mechanistic interpretation. In particular, no-causal should not be described as removing head discovery, and no-morph should not be described as deleting the morphology head: its reported parameter count is unchanged.

Within each pair, all seven rows have the same reported token count, update count, and effective batch size. EN-FR uses 7,854,360 tokens and 790 updates; EN-FA uses 7,341,738 tokens and 780 updates; EN-ZH uses 8,399,100 tokens and 790 updates. The effective batch size is 128 throughout. These fields support workload comparability within the exports, while leaving checkpoint provenance, seed alignment, and configuration matching to be verified.

B.2 Translation quality

Table 1 reports the primary quality values without altering their means or standard deviations. Full-CRAT has the highest displayed mean in all nine language-pair and metric combinations. Figure 1 shows decreases calculated as the displayed full-model mean minus the ablated-model mean. All differences are absolute score differences; COMET remains on its original numerical scale.

Table 1. Translation quality for all ablations. Values are reported mean ± SD.

Variant	BLEU	chrF	COMET
EN-FR			
Full-CRAT	**31.358 ± 0.096**	**55.763 ± 0.108**	**0.80679 ± 0.00098**
no-kl	30.841 ± 0.433	55.248 ± 0.281	0.80163 ± 0.00134
no-causal	31.129 ± 0.417	55.202 ± 0.306	0.80420 ± 0.00081
no-morph	31.100 ± 0.250	55.264 ± 0.303	0.80444 ± 0.00108
no-sparse	31.142 ± 0.272	55.244 ± 0.407	0.80397 ± 0.00087
no-integration	31.209 ± 0.445	55.262 ± 0.285	0.80403 ± 0.00139
static-role	30.642 ± 0.152	54.900 ± 0.233	0.80180 ± 0.00086

EN-FA			
Full-CRAT	**8.685 ± 0.127**	**27.176 ± 0.143**	**0.67259 ± 0.00541**
no-kl	7.884 ± 0.241	25.924 ± 0.389	0.65346 ± 0.00224
no-causal	8.187 ± 0.408	26.136 ± 0.559	0.65097 ± 0.00083
no-morph	8.359 ± 0.300	26.014 ± 0.543	0.65139 ± 0.00217
no-sparse	8.494 ± 0.527	26.143 ± 0.755	0.65247 ± 0.00087
no-integration	7.980 ± 0.372	25.674 ± 0.582	0.64881 ± 0.00291
static-role	7.813 ± 0.279	25.802 ± 0.378	0.66076 ± 0.00078
EN-ZH			
Full-CRAT	**26.984 ± 0.247**	**43.746 ± 0.014**	**0.80757 ± 0.00203**
no-kl	24.272 ± 0.875	43.189 ± 0.234	0.80174 ± 0.00253
no-causal	24.185 ± 1.133	43.377 ± 0.093	0.80317 ± 0.00298
no-morph	23.843 ± 1.442	43.074 ± 0.116	0.80122 ± 0.00123
no-sparse	24.268 ± 1.408	43.397 ± 0.168	0.80204 ± 0.00288
no-integration	24.243 ± 0.840	43.301 ± 0.081	0.80211 ± 0.00278
static-role	24.034 ± 1.440	43.168 ± 0.135	0.80150 ± 0.00270

Higher is better. Bold values identify the highest displayed mean within each pair and metric. SD is the reported standard deviation. The full-model reference discrepancy in Appendix A applies to all comparisons with Full-CRAT.

For EN-FR, the differences are comparatively small. Relative to Full-CRAT, no-integration has the smallest BLEU decrease, 0.148, while no-morph has the smallest chrF and COMET decreases, 0.499 and 0.00235. Static routing has the largest BLEU and chrF decreases, 0.715 and 0.862, whereas no-kl has the largest COMET decrease, 0.00517.

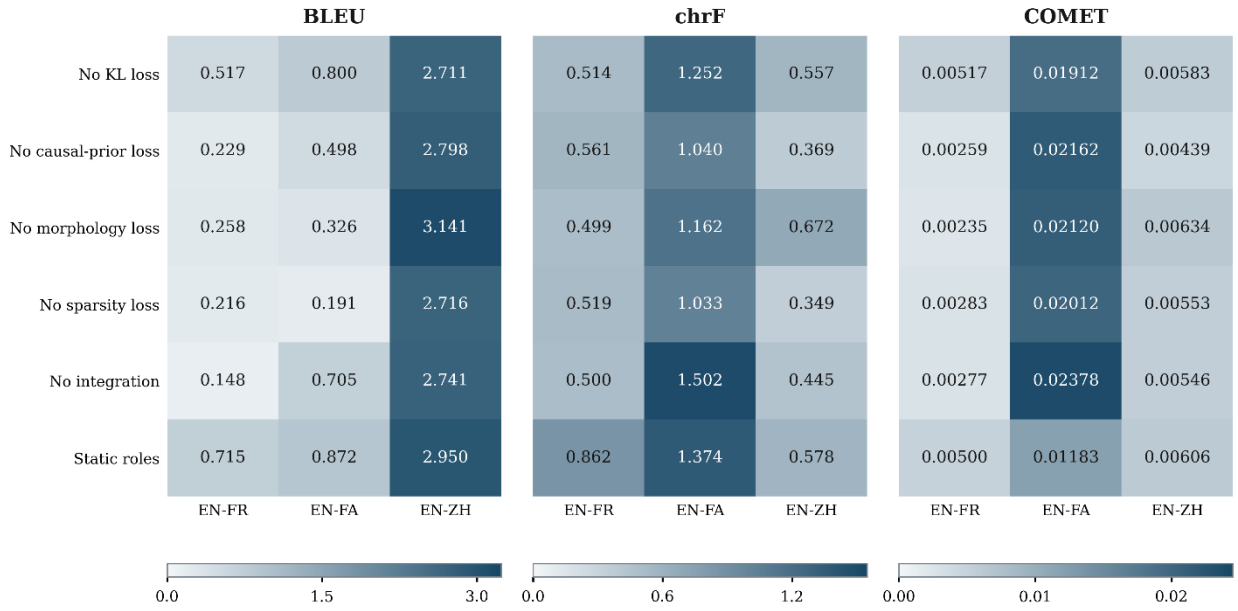


Figure 1. Absolute quality decreases relative to the displayed Full-CRAT means. Each metric uses its own color scale; darker cells indicate a larger decrease. All 54 differences are recomputed from the primary means.

Among the ablated variants, the chrF difference between no-morph and no-integration is only 0.00143. This small separation should not be used to claim a reliable ordering without paired observations.

For EN-FA, removing integration produces the largest chrF and COMET decreases, 1.502 and 0.02378, together with a 0.705 BLEU decrease. Static routing has the largest BLEU decrease, 0.872, and a 1.374 chrF decrease. Its COMET decrease is only 0.01183, however, which is the smallest COMET decrease among the ablations. This divergence is visible directly in the ablated means: static-role has the lowest BLEU but the highest COMET among the six variants. It therefore cannot be characterized as uniformly worst for translation quality. The no-sparse variant retains the highest ablated BLEU and chrF, at 8.494 and 26.143, but its COMET value of 0.65247 remains below static-role.

For EN-ZH, the displayed BLEU decreases range from 2.711 to 3.141, equivalent to 10.05-11.64% of the displayed Full-CRAT BLEU. The no-morph variant is lowest on all three quality metrics, with decreases of 3.141 BLEU, 0.672 chrF, and 0.00634 COMET. The best ablated result depends on the metric: no-kl has the highest BLEU, no-sparse has the highest chrF, and no-causal has the highest COMET. The BLEU difference between no-kl and no-sparse is only 0.00430. The large common BLEU offset against Full-CRAT is especially sensitive to the reference mismatch: the source delta columns imply a full-model BLEU near 24.254, compared with the displayed 26.984.

Across variants, the mean decrease within each pair is 0.347 BLEU, 0.576 chrF, and 0.00345 COMET for EN-FR; 0.566, 1.227, and 0.01961 for EN-FA; and 2.843, 0.495, and 0.00560 for EN-ZH. These are descriptive averages over six ablations, not pooled test-set scores. They locate the main sensitivities without combining metrics that have different scales or treating the language pairs as interchangeable evaluation samples.

B.3 Contributions of the individual components

KL regularization. The no-kl variant decreases BLEU by 0.517, 0.800, and 2.711 on EN-FR, EN-FA, and EN-ZH, respectively. The corresponding COMET decreases are 0.00517, 0.01912, and 0.00583. Its substantial memory and training-time savings make it a meaningful quality-cost trade-off. In the earlier implementation, KL regularization uses a frozen teacher, so teacher removal is a plausible explanation for these savings. The table alone does not confirm the teacher lifecycle in the ablation runs. On EN-ZH, no-kl also reduces the evaluation soft-gate count above 0.5 from 2.5 to 2.0; this change describes the logged gates rather than actual top-k computation.

Causal-prior alignment. The no-causal variant decreases BLEU by 0.229, 0.498, and 2.798, with COMET decreases of 0.00259, 0.02162, and 0.00439. Its most distinctive effect is a redistribution of gate magnitudes: the training indicator gate increases in every pair while the integration gate decreases. This behavior is consistent with the intended role of the prior-alignment objective in constraining routing. It does not validate causal head identification or show that head discovery was removed. Among EN-ZH ablations, no-causal retains the highest COMET, further showing that routing changes and quality changes do not admit one universal ranking.

Morphology supervision. The no-morph variant decreases BLEU by 0.258, 0.326, and 3.141, with COMET decreases of 0.00235, 0.02120, and 0.00634. EN-ZH has the lowest ablated quality under this variant on every metric, whereas EN-FR retains the highest ablated chrF and COMET. This pattern supports a pair-dependent role for the auxiliary objective, conditional on matched references. It does not establish that the effect scales with morphological richness: the largest BLEU decrease occurs on EN-ZH, and aggregate translation metrics do not isolate morphological errors. The nearly unchanged mean gates suggest that any effect is not explained by a large shift in average routing alone.

Sparsity regularization. The no-sparse variant decreases BLEU by 0.216, 0.191, and 2.716, with COMET decreases of 0.00283, 0.02012, and 0.00553. It has the smallest unweighted average BLEU decrease across pairs, 1.041, and the smallest average chrF decrease, 0.634. The soft gates increase slightly, as expected when a mean-gate penalty is removed, but the number of gates exceeding 0.5 is unchanged in every pair. The measured training-time changes have mixed signs. These observations support a modest effect on gate magnitudes; they do not demonstrate fewer executed roles or a consistent speed advantage from the sparsity term.

Integration adapters. The no-integration variant decreases BLEU by 0.148, 0.705, and 2.741, with COMET decreases of 0.00277, 0.02378, and 0.00546. Its largest chrF and COMET losses occur on EN-FA. Removing 393,216

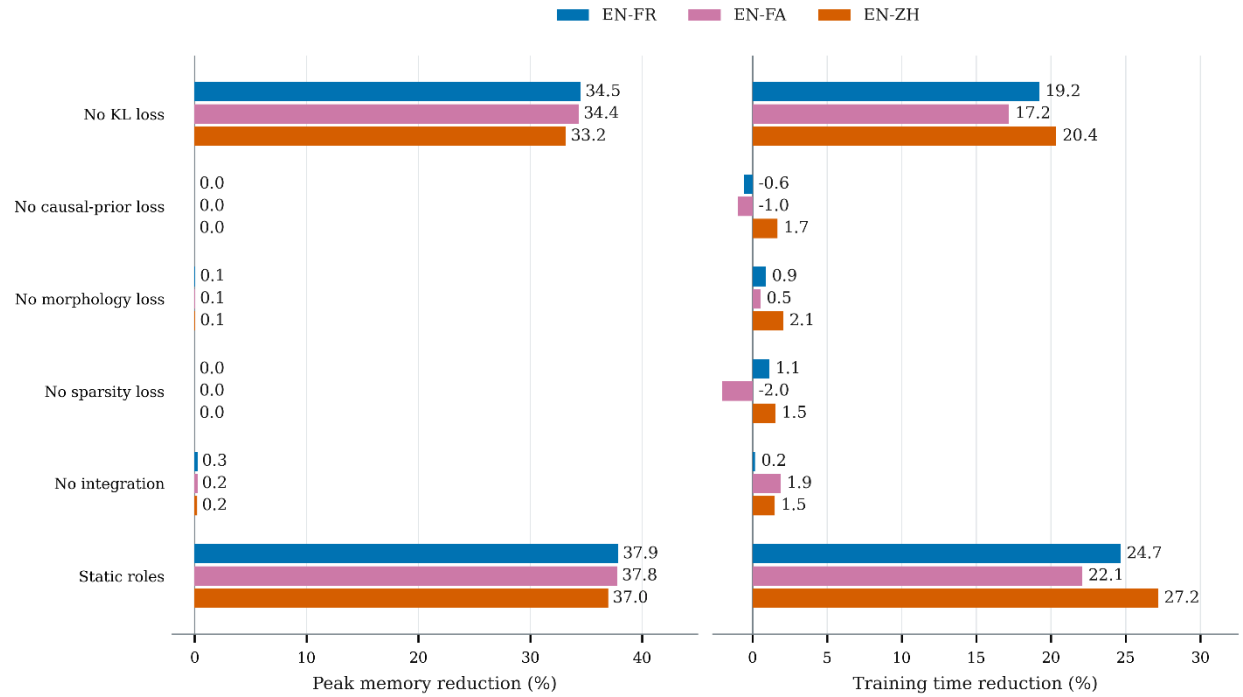


Figure 2. Peak-memory and training-time reductions relative to the displayed Full-CRAT resource values. Positive values indicate savings, and negative values indicate increased cost. Percentages use the primary measurements, with matched reported token counts within each pair.

trainable parameters saves only about 0.100 GiB of peak memory per pair and 0.17-1.89% of training time. Against the displayed scores, retaining integration therefore has a small measured resource cost and an appreciable quality association on EN-FA. A nonzero integration gate remains in the routing export; that gate does not establish that an integration adapter was executed.

Static roles. The static-role variant decreases BLEU by 0.715, 0.872, and 2.950, with COMET decreases of 0.00500, 0.01183, and 0.00606. It gives the largest average BLEU and chrF decreases across pairs, 1.512 and 0.938, but the smallest average COMET decrease, 0.00763. Its parameter reduction is exactly 4,295,469 in every pair. Under the earlier architecture dimensions, this equals the router plus morphology-head parameter counts, so static-role is consistent with a compound architecture change. Its quality and resource differences should not be attributed exclusively to the use of static gates.

B.4 Training cost and parameter efficiency

The no-kl, no-causal, no-morph, and no-sparse variants retain exactly the Full-CRAT trainable parameter count. The no-integration reduction is 0.201%, 0.181%, and 0.163% for EN-FR, EN-FA, and EN-ZH. Static-role reduces trainable parameters by 2.197%, 1.974%, and 1.777%. These differences are small relative to the parameter budget of the tuned head slices. The much larger memory savings in no-kl and static-role consequently reflect changes beyond trainable parameter count alone.

Disabling KL regularization reduces peak memory by 34.51%, 34.35%, and 33.20% across EN-FR, EN-FA, and EN-ZH. The absolute saving is 13.69-13.78 GiB. Training time decreases by 19.21%, 17.19%, and 20.36%, corresponding to 10.63-14.16 minutes per reported run. Since the token counts are matched within each pair, the corresponding training-throughput increases of 23.77%, 20.75%, and 25.56% are consistent with faster processing of the same reported workload.

Static-role gives the lowest peak memory and shortest training time in all three pairs. Its peak-memory savings are 37.87%, 37.77%, and 36.95%, and its training-time savings are 24.66%, 22.08%, and 27.17%. These savings are

operationally substantial, but the approximately 15 GiB memory difference cannot be explained solely by the removal of 4.30 million trainable parameters. The complete configuration of auxiliary objectives, teacher loading, and forward-pass requirements must be considered alongside the routing change.

For the remaining variants, peak-memory differences are negligible or small, and training-time changes lie between a 2.04% increase and a 2.06% reduction. No-causal is slightly slower on EN-FR and EN-FA, and no-sparse is slower on EN-FA. Without repeated timing measurements and matched hardware conditions, these small changes do not establish reliable speed effects. The resource percentages themselves reconcile with the displayed primary resource values within numerical rounding, unlike the supplied quality deltas.

Table 2. Trainable parameters and training resource measurements.

Variant	Trainable (M)	Trainable (%)	Peak (GiB)	Training (min)	Training tokens/s	Time saved (%)
EN-FR						
Full-CRAT	195.530	2.629	39.693	64.26	2,037.1	+0.00
no-kl	195.530	2.629	25.996	51.92	2,521.4	+19.21
no-causal	195.530	2.629	39.693	64.62	2,025.8	-0.56
no-morph	195.530	2.629	39.670	63.69	2,055.4	+0.89
no-sparse	195.530	2.629	39.693	63.53	2,060.4	+1.13
no-integration	195.136	2.624	39.593	64.15	2,040.6	+0.17
static-role	191.234	2.573	24.663	48.42	2,703.8	+24.66
EN-FA						
Full-CRAT	217.550	2.916	40.102	61.84	1,978.6	+0.00
no-kl	217.550	2.916	26.327	51.21	2,389.2	+17.19
no-causal	217.550	2.916	40.102	62.44	1,959.7	-0.96
no-morph	217.550	2.916	40.079	61.52	1,989.1	+0.53
no-sparse	217.550	2.916	40.102	63.11	1,938.9	-2.04
no-integration	217.156	2.911	40.002	60.68	2,016.6	+1.89
static-role	213.254	2.861	24.954	48.19	2,539.2	+22.08
EN-ZH						
Full-CRAT	241.667	3.229	41.238	69.53	2,013.4	+0.00
no-kl	241.667	3.229	27.547	55.37	2,528.1	+20.36
no-causal	241.667	3.229	41.238	68.36	2,047.9	+1.68
no-morph	241.667	3.229	41.215	68.10	2,055.7	+2.06
no-sparse	241.667	3.229	41.238	68.46	2,044.8	+1.54
no-integration	241.274	3.224	41.138	68.48	2,044.2	+1.51
static-role	237.371	3.174	25.999	50.64	2,764.4	+27.17

M denotes millions. Trainable percentage uses the registered total parameter count of each variant. Positive time savings indicate a shorter duration relative to the displayed Full-CRAT resource row; negative values indicate a longer duration.

B.5 Generation efficiency and latency

Static-role has the highest batched generation throughput in every pair, exceeding Full-CRAT by 4.70%, 7.28%, and 4.33%. Its mean single-example latency is also lower by 5.99%, 6.14%, and 2.68%. These improvements do not extend

uniformly to the latency tail. On EN-ZH, the reported 95th-percentile latency increases from 5.079 s to 11.895 s for static-role, a 134.18% increase; no-kl reaches 10.286 s, a 102.51% increase. EN-FA static-role also has a slightly higher 95th percentile, 9.018 s versus 8.810 s. Thus, lower average latency does not establish better tail behavior.

Generation duration also depends on output length. For example, EN-FA static-role generates 27,466 tokens, compared with 28,544 for Full-CRAT, so its 10.31% shorter total generation time overstates its 7.28% token-throughput improvement. Conversely, EN-FA no-integration takes 0.98% longer overall but has 1.47% higher token throughput because it generates more tokens. In the earlier implementation, the timer starts after tokenization, feature construction, and router computation. The reported generation timings therefore do not measure complete request latency or isolate routing overhead. The separate single-example benchmark should also be kept distinct from batched evaluation throughput.

Table 3. Generation duration, output volume, throughput, and single-example latency.

Variant	Gen. (s)	Output tokens	Batch tokens/s	Mean (s)	p50 (s)	p95 (s)	Single tokens/s
EN-FR							
Full-CRAT	521.5	27,172	52.10	1.728	0.848	3.502	13.75
no-kl	521.2	27,116	52.02	1.709	0.838	3.276	13.75
no-causal	524.3	27,192	51.86	1.703	0.845	3.238	13.94
no-morph	519.6	27,154	52.26	1.693	0.856	3.204	14.10
no-sparse	519.8	27,130	52.19	1.728	0.845	3.484	13.77
no-integration	512.4	27,138	52.96	1.701	0.846	3.458	13.84
static-role	494.7	26,988	54.55	1.624	0.805	2.949	14.15
EN-FA							
Full-CRAT	603.8	28,544	47.27	1.917	0.910	8.810	13.94
no-kl	581.9	28,514	49.01	1.848	0.946	6.811	13.83
no-causal	601.6	28,846	47.95	1.939	0.904	9.247	14.35
no-morph	579.8	28,875	49.80	1.934	0.917	8.855	14.10
no-sparse	569.6	28,822	50.60	1.881	0.910	9.445	13.84
no-integration	609.7	29,246	47.96	1.921	0.901	9.426	14.25
static-role	541.6	27,466	50.71	1.800	0.882	9.018	14.06
EN-ZH							
Full-CRAT	564.9	31,507	55.78	2.806	1.549	5.079	12.60
no-kl	575.4	31,446	54.65	2.911	1.566	10.286	12.65
no-causal	576.1	31,828	55.24	2.813	1.550	5.063	12.77
no-morph	569.7	31,454	55.21	2.790	1.512	5.050	12.44
no-sparse	575.8	31,774	55.18	2.842	1.520	5.060	12.59
no-integration	570.4	31,733	55.63	2.812	1.503	5.087	12.63
static-role	543.1	31,602	58.19	2.730	1.456	11.895	12.81

Batch tokens/s comes from the complete evaluation generation workload. Mean, p50, p95, and Single tokens/s come from the separate single-example latency benchmark in the supplied implementation. p50 and p95 denote the reported 50th and 95th percentiles. Lower duration and latency, and higher throughput, are preferable.

B.6 Routing behavior

Full-CRAT assigns the largest evaluation mean gates to position and source on EN-FR and EN-FA, and to source and position on EN-ZH. The corresponding source/position values are 0.814/0.880, 0.802/0.850, and 0.922/0.794. Integration is stronger on EN-FR and EN-FA, at 0.642 and 0.647, than on EN-ZH, at 0.443. These are averages of independent sigmoid gates. Their values do not sum to one and should not be reported as categorical role probabilities.

Removing causal-prior alignment increases the training indicator gate by 0.120, 0.118, and 0.193 while reducing the integration gate by 0.307, 0.209, and 0.261. At evaluation, the integration decrease persists, at 0.068, 0.093, and 0.146, but the indicator response differs by pair: it decreases by 0.029 on EN-FR and increases by 0.070 and 0.049 on EN-FA and EN-ZH. Training snapshots and evaluation averages summarize different data and model states, so their differences should not be interpreted as a direct measure of generalization or calibration.

Table 4. Evaluation mean soft gates and diagnostic threshold counts.

Variant	Source	Indicator	Position	Integration	Active fractions (S / I / P / G)	Mean count above 0.5
EN-FR						
Full-CRAT	0.8144	0.4271	0.8798	0.6421	1 / 0 / 1 / 1	3.0
no-kl	0.8190	0.4265	0.8825	0.6423	1 / 0 / 1 / 1	3.0
no-causal	0.8023	0.3985	0.8729	0.5745	1 / 0 / 1 / 1	3.0
no-morph	0.8141	0.4266	0.8797	0.6421	1 / 0 / 1 / 1	3.0
no-sparse	0.8185	0.4361	0.8821	0.6511	1 / 0 / 1 / 1	3.0
no-integration	0.8143	0.4272	0.8797	0.6421	1 / 0 / 1 / 1	3.0
EN-FA						
Full-CRAT	0.8016	0.4386	0.8502	0.6465	1 / 0 / 1 / 1	3.0
no-kl	0.8019	0.4342	0.8522	0.6471	1 / 0 / 1 / 1	3.0
no-causal	0.8146	0.5083	0.8553	0.5536	1 / 1 / 1 / 1	4.0
no-morph	0.8016	0.4381	0.8500	0.6461	1 / 0 / 1 / 1	3.0
no-sparse	0.8055	0.4472	0.8531	0.6490	1 / 0 / 1 / 1	3.0
no-integration	0.8014	0.4389	0.8501	0.6450	1 / 0 / 1 / 1	3.0
EN-ZH						
Full-CRAT	0.9223	0.4764	0.7938	0.4428	1 / 0.5 / 1 / 0	2.5
no-kl	0.8979	0.4034	0.7417	0.4399	1 / 0 / 1 / 0	2.0
no-causal	0.9319	0.5250	0.8007	0.2970	1 / 1 / 1 / 0	3.0
no-morph	0.9222	0.4775	0.7946	0.4429	1 / 0.5 / 1 / 0	2.5
no-sparse	0.9262	0.4950	0.8030	0.4521	1 / 0.5 / 1 / 0	2.5
no-integration	0.9223	0.4767	0.7944	0.4420	1 / 0.5 / 1 / 0	2.5

S, I, P, and G identify source, indicator, position, and integration. Active fractions are the reported proportions of soft gates strictly greater than 0.5, in that order.

The soft-threshold counts in Table 4 increase from 3 to 4 on EN-FA and from 2.5 to 3 on EN-ZH under no-causal. Under no-kl, EN-ZH decreases from 2.5 to 2; the other pairs remain at 3. Removing sparsity regularization increases every evaluation mean gate by approximately 0.002-0.019 but leaves the threshold counts unchanged. No-morph and no-integration change each evaluation mean gate by less than 0.0016. These small changes in marginal means do not rule out changes in per-example rankings or representations.

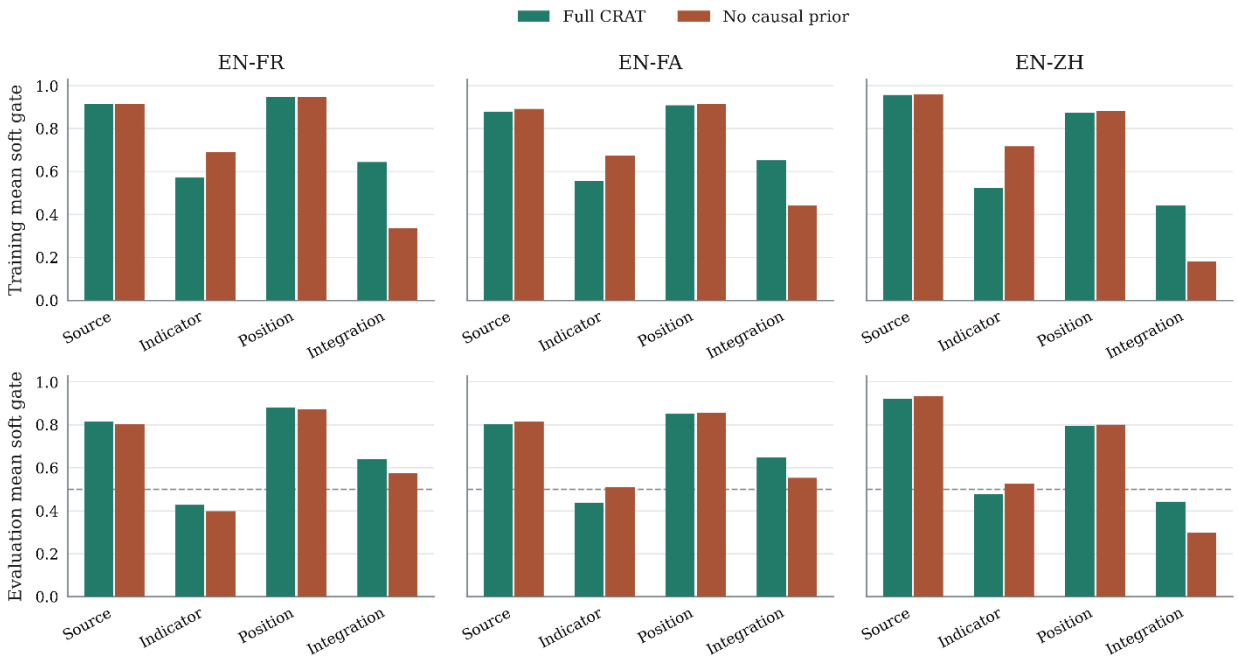


Figure 3. Mean soft gates for Full-CRAT and no-causal during training and evaluation. The dashed evaluation line marks the diagnostic threshold of 0.5. These are independent sigmoid gates recorded before evaluation top-k masking.

The logged threshold count is the mean number of soft gates strictly greater than 0.5. The earlier code returns these soft gates for logging while separately masking them for top-k evaluation, with a default of two selected roles. Accordingly, counts of 2.5, 3, or 4 do not mean that those numbers of roles were executed. Nor can the most frequently selected pair of roles be recovered from mean gates. The missing static-role routing entries mean that these diagnostics were not reported; they are not zero gates or zero role usage.

B.7 Variability and interpretation

The displayed Full-CRAT standard deviations are smaller than those of every ablation for BLEU and chrF in all three pairs, covering 36 pair-metric-variant comparisons. COMET does not follow this pattern. Full-CRAT has a smaller COMET standard deviation in only eight of the 18 comparisons. On EN-FA, its COMET standard deviation is 0.00541, compared with 0.00078-0.00291 for the ablations. The results therefore do not support a claim that the full method is uniformly more stable across metrics. These comparisons also inherit the full-reference provenance issue.

The exports report means and columns labeled as standard deviations, but do not encode seed identifiers, the seed count, or paired run-level measurements. The earlier main-comparison description specified six seeds; this does not independently establish the aggregation protocol for the ablation files. No confidence intervals, paired significance tests, or significance markers are inferred from these summaries. The reported resource measurements likewise lack variability estimates. Differences between means and overlaps between standard-deviation ranges do not, by themselves, settle statistical significance.

The component ranking is consequently metric dependent. Under the displayed reference, morphology removal has the largest EN-ZH quality losses, integration removal has the largest EN-FA chrF and COMET losses, and static roles have the largest EN-FR BLEU and chrF losses. Among the ablated rows, these orderings do not depend on which full-model reference is chosen. However, the claim that every component improves every metric does depend on that reference: five supplied deltas change sign when recomputed against the displayed full-model means.

The training-resource evidence supports a clear distinction between expensive and inexpensive modifications. No-kl and static-role substantially reduce peak memory and training time, whereas the other variants have small measured resource effects. The routing evidence associates removal of the causal-prior objective with the largest average gate

redistribution. These findings can guide interpretation of CRAT's design without asserting that the components are independent or additive. Single-removal comparisons do not establish interaction effects, and static-role appears to change more than the router alone.

The displayed scores are consistent with a quality advantage for the complete CRAT configuration, with the largest BLEU sensitivity on EN-ZH and COMET sensitivity on EN-FA. That advantage remains a descriptive result until the full-model references, ablation configurations, and seed-level records are reconciled. A final experimental claim should retain the observed metric-specific trade-offs, distinguish training savings from generation latency, and describe the routing diagnostics according to what the implementation actually logs.

Appendix A Reference reconciliation

For each metric and ablated row, subtracting the supplied delta from the reported ablated mean reconstructs the reference used by that delta. Table A1 compares those implied references with the displayed Full-CRAT values. The implied references agree across variants to export precision, apart from a sub-millionth COMET discrepancy on EN-ZH. Their consistency suggests a reference-version mismatch, although the exports do not establish which reference corresponds to the ablation checkpoints. The implied values are diagnostic reconstructions, not verified replacement results.

Table A1. Displayed and implied Full-CRAT quality references.

Pair	Metric	Displayed full	Implied reference	Displayed minus implied
EN-FR	BLEU	31.357525	31.198672	+0.158854
EN-FR	chrF	55.762685	55.279893	+0.482792
EN-FR	COMET	0.806794	0.804601	+0.002193
EN-FA	BLEU	8.684953	8.582031	+0.102922
EN-FA	chrF	27.176113	26.455343	+0.720770
EN-FA	COMET	0.672590	0.654790	+0.017799
EN-ZH	BLEU	26.983584	24.253650	+2.729934
EN-ZH	chrF	43.746007	43.451922	+0.294086
EN-ZH	COMET	0.807566	0.802352	+0.005214

Implied reference is the mean across the six reconstructed reference values. It is presented only to diagnose the mismatch. No replacement of the displayed primary means has been made.

Table A2 lists the five sign reversals; the companion audit table retains every supplied and recomputed value. In contrast, resource percentages, parameter fractions, and token-rate identities reconcile within approximately 0.000001 in their respective units.

Table A2. Quality comparisons whose sign changes after recomputation.

Pair	Variant	Metric	Supplied delta	Recomputed delta
EN-FR	no-integration	BLEU	+0.010695	-0.148159
EN-FA	static-role	COMET	+0.005974	-0.011825
EN-ZH	no-kl	BLEU	+0.018535	-2.711399
EN-ZH	no-causal	COMET	+0.000822	-0.004392
EN-ZH	no-sparse	BLEU	+0.014237	-2.715697

Here both deltas use ablated mean minus full-model mean, matching the source delta-column convention. Positive values favor the ablation. Figure 1 instead shows score decreases, using the opposite sign.

All 18 displayed full-model quality means and standard deviations match the earlier Mistral results exactly at the supplied precision. However, the earlier Mistral CRAT rows have 316/312/316 updates for EN-FR/EN-FA/EN-ZH, compared with 790/780/790 here. Their token counts are 3,141,744/2,937,945/3,359,640, compared with 7,854,360/7,341,738/8,399,100 here.

Appendix B Training routing diagnostics

Table B1. Training mean soft gates and the logged gate statistic.

Variant	Source	Indicator	Position	Integration	Logged gate statistic
EN-FR					
Full-CRAT	0.9157	0.5717	0.9457	0.6439	0.72906
no-kl	0.9149	0.5762	0.9444	0.6454	0.72817
no-causal	0.9136	0.6915	0.9463	0.3369	0.70425
no-morph	0.9154	0.5720	0.9457	0.6440	0.72911
no-sparse	0.9185	0.5818	0.9470	0.6531	0.71544
no-integration	0.9157	0.5718	0.9456	0.6426	0.72981
EN-FA					
Full-CRAT	0.8793	0.5572	0.9083	0.6527	0.78015
no-kl	0.8773	0.5471	0.9078	0.6540	0.78814
no-causal	0.8923	0.6750	0.9148	0.4437	0.77401
no-morph	0.8795	0.5570	0.9084	0.6525	0.78007
no-sparse	0.8811	0.5639	0.9094	0.6596	0.77129
no-integration	0.8790	0.5566	0.9082	0.6496	0.78218
EN-ZH					
Full-CRAT	0.9569	0.5245	0.8756	0.4424	0.84810
no-kl	0.9565	0.5316	0.8763	0.4429	0.84565
no-causal	0.9592	0.7178	0.8827	0.1811	0.64079
no-morph	0.9567	0.5243	0.8758	0.4424	0.84800
no-sparse	0.9576	0.5356	0.8792	0.4519	0.83834
no-integration	0.9568	0.5233	0.8757	0.4418	0.84854